



STRUCTURAL ENGINEERING - OCEAN TRANSPORTATION ANALYSIS FOR MODULES

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1.0 PURPOSE

This guideline establishes the recommended procedures for the design of structural modules to withstand inertial forces imposed by probable extreme vessel motions during ocean transport.

Increasingly modules are assembled anywhere in the world, including on continents different than the project site location, and then are transported by seagoing vessel to the site. The intent of this guideline is to provide a consistent basis for the design of modules subjected to ocean transportation loads.

2.0 APPLICATION

This guideline is used by Structural Engineers on all projects with modular execution as a resource during the design of equipment, piperack, and building modules transported over water.

Project specific design requirements vary widely between projects based on client preference, local codes, marine warranty surveyor (MWS) requirements to approve design for transportation, and many other factors. This guideline covers the fundamental design considerations of these modules, and not every potential detail that may be required.

In general, ocean transport of modules using barges and cargo vessels is more severe than land transport using self-propelled modular transports (SPMT), trucks, or trains. If land transport applies to the project, instead of or in addition to ocean transport, this needs to be assessed separately. Modules that are lifted as part of the transportation sequence also must be assessed for this scenario separately.

3.0 INTERFACING DISCIPLINES

- Construction
- Electrical Engineering
- Mechanical Engineering
- Naval Architect
- Piping Stress Engineering
- Supply Chain

4.0 DEFINITIONS

Barge

Flat-bottomed vessels used to transport modules across seas and inland waters such as intracoastal waterways. Barges are not self-propelled. They are typically towed using tugs both for inland and oceanic voyages. Ocean going "push-barges" are generally bulk cargo or tankers and are not generally used for transporting modules or tall cargoes. On such vessels push tugs mate into the aft notch of the barge. Inland push tugs push the barges at vertical transom structures or at skegs. Barges are directionally more stable when towed from bow than when pushed from stern.



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Center of Gravity (CoG)	A point at which the mass of the body or system is assumed to be concentrated at.
Center of Rotation (CoR)	Vessel center of rotation is at the center of floatation, which is mostly the same horizontal location as the vessel deck center.
Grillage	Set of arranged temporary steel on the deck of the ship to spread vertical loads (and lateral loads where applicable) from the module columns onto hull structure. The grillage assembly are necessary components to support the module during set-down and are designed for the full set-down load. Together with the sea fastening components (which are installed after set-down) they comprise the temporary structural elements to support modular cargo during its voyage. Grillages should have enough transverse breaks along the hull to ensure that they do not contribute to the hull girder stiffness. If they exceed 40% of length of the hull, they could become part of hull longitudinal girder and attract significant hog and sag loading during the tow. Grillage are bespoke structures unique to the hull and module involved in the tow. Unless the tow configuration is repeated multiple times these structures are used only once. Hence, they do not require coatings.
Heave	Vessel vertical movement
Hog and Sag	Longitudinal bending of the deck resulting from ship flexing due to interaction with waves. The magnitude of these bending moments is a function of ratio of wave length to the waterline length of the hull. The moments are maximum when that ratio is close to one.
Lift-on/Lift-off	The process of loading or unloading a module from a vessel using cranes.
Loadout	The operation of transferring a module from the place of fabrication to a vessel, to be further transported on the vessel to the final destination.
Longitudinal Direction	The direction parallel to the center line along hull from (stern) aft to (bow) forward. Refer to Figure 1b.
Marine Warranty Surveyor	A marine warranty surveyor (MWS) provides independent third-party technical review and approval of high-value and high-risk marine construction and transportation project operations, from the planning stages through to the physical execution. The MWS provides insurance for the transportation of a module from the time the complete module is lifted off its supports (by crane or by SPMT) at the module yard until the time it is set down on its foundation at site. For marine transportation, land transportation, and lifting, the authority having jurisdiction will be the marine warranty surveyor and their approval of the



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	whole transportation analysis must be received prior to shipment. Refer to Attachment 01 for an example of MWS activities during a project execution. Each project should develop an execution plan depending on the project size and schedule.
MetOcean Data	Provides estimated environmental conditions during ocean transport including wind speed, surface current, wave height and direction, air temperature, humidity and expected sea level changes, etc.
Module	Equipment, piping, electrical items, and other components that are pre-assembled in structures at a remote assembly site and then transported to the project site using road, rail, or water transportation methods.
Naval Architect	A specialized engineering discipline that determines marine transportation motion, accelerations, and stability based on vessel size, voyage duration, route, season, module weight and vessel characteristics.
Off-load	The operation of transferring a transported module from the vessel to the final destination.
Pitch	Vessel angular rotation about transverse axis.
Roll	Vessel angular rotation about longitudinal axis.
Roll-on/Roll-off	The process of loading or unloading a module from a vessel using SPMTs.
Sea Fastening	Connecting members and material used in combination with grillage to transfer loading from the module and grillage to the frame of the ship. Sea fastening is installed after the module is set down on the deck, and includes wing plates on grillage, pitch stoppers, roll stoppers, roll braces, pitch braces, and connecting gusset plates. Sea-fastening is intended to be the primary load path for lateral loads that are generated during marine transportation including minor lateral loads during harbor conditions.
Ship	Sea / ocean-going vessel used to transport modules from the fabrication yard to site.
SPMT	Self-Propelled Modular Transporter. A heavy haul trailer with multiple axles used to transport modules on land over short distances, such as between the module fabrication yard and the vessel.
Surge	Vessel translation along longitudinal axis
Sway	Vessel translation along transverse axis
Transverse Direction	The direction perpendicular to the center line of the transport vessel Refer to Figure 1b.
Yaw	Vessel angular rotation about vertical axis.



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Web Frame

Structural components of a ship consisting of steel frames oriented transverse to the ship length and directly supporting the ship deck, spaced at regular intervals. Web frames are the primary load transferring mechanism used to support loading from the sea fastening and grillage. In hull terminology they are transverse deck frames. There are also deck plating, longitudinal frames and longitudinal girders that are significant hull structural components relevant to module transportation structural analysis.

Weight Control

The management of a module's weight and CoG to enable the module to meet the logistics requirements for transportation. Refer to Fluor Guideline 000.200.0880 (Weight Control Management) for more detail.

5.0 REQUIREMENTS

Not Used

6.0 EXECUTION OF GUIDELINES

This guideline contains the following major sections:

- Introduction – an overview of ocean transportation.
- Transport Vessel Types and Motions – as vessel rolls, pitches and heaves in the ocean, inertial acceleration forces will be generated which are different for each type of vessel.
- Applicable Codes (for Marine Warranty Surveyor approval).
- Design Ocean Acceleration – a description of the formulae to calculate ocean inertia forces on transported modules.
- Design Margins
- Ocean Wind – a reference that could be used for application of ocean wind.
- Hog & Sag – a definition of vessel hog & sag and its loading imposed on a transported module.
- Structural Model Implementation – a step-by-step analysis to apply ocean inertia forces on transported modules with implementation of center of gravity (COG) shift/tolerance. A sample design is included using RISA-3D as an aid in producing actual design.
- Lift-on/lift-off, roll-on/roll-off, grillage and sea fastening and fatigue design – a discussion and references that address design considerations for lift-on/lift-off, load-in/load-out (SPMT transportation), grillage and sea fastening and design for fatigue.

6.1 Introduction

Modules are required to be designed to withstand the loads caused by the most adverse environmental conditions expected for the area and season through which it will be transported. For ocean transportation the design criteria should be defined, consisting of the design wave and design wind, based on the MetOcean data which is applicable for the planned navigation route



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from the module fabrication yards to site. Typically MetOcean specialists provide the MetOcean data report, however it may be available through project specifications as well. Clients often hire a MetOcean consultant on major projects. The report will have data for in-place conditions and for the transportation route. MetOcean data for in-place conditions are applicable only when modules are installed at offshore locations).

During ocean transportation, the module will be subjected to inertial loads due to the rigid-body motions of the transportation vessels in the sea state and the forced deflections of its supports due to the flexural response of the vessel to waves in hog and sag conditions.

Forces acting on a vessel are constantly varying in direction and frequency. A vessel is free to move about six degrees of freedom. There are three translation movements (surge, sway, heave), and three rotation movements (roll, pitch, yaw), as shown in Figure 1a. These movements introduce dynamic forces that result in stresses in the vessel's structure and any cargo on board. Modules must be designed to accommodate the dynamic forces generated during ocean transportation.

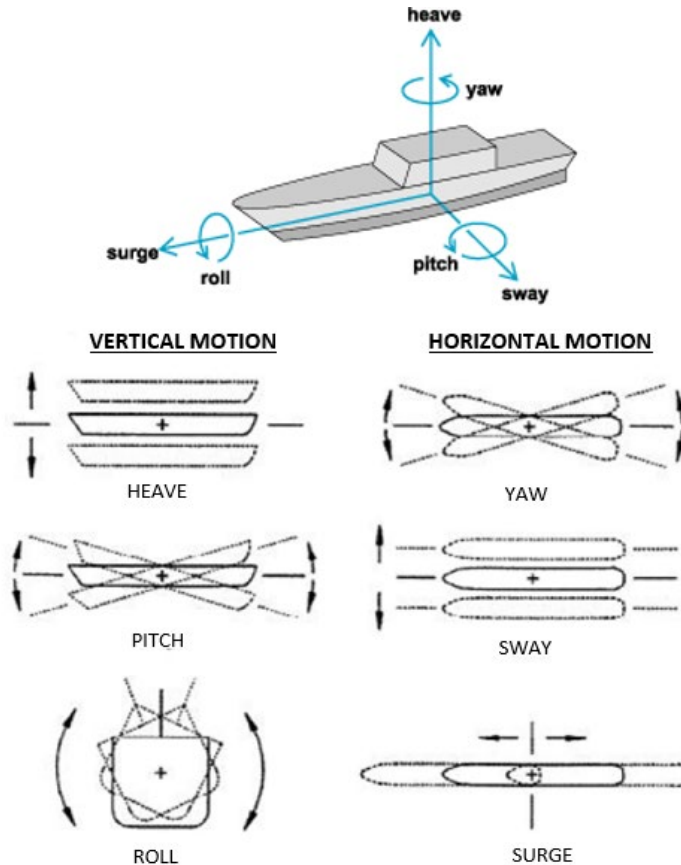


Figure 1A – VESSEL MOTIONS



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Direction of the sea surface motion, in relation to the vessel heading, is expressed in three different conditions: head sea, beam sea, and quartering sea, as shown in Figure 1b.

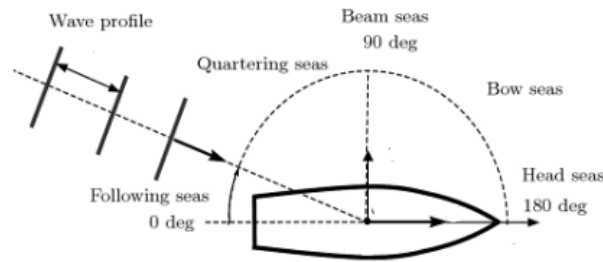


Figure 1B – Vessel Heading Angles

6.2 Transportation Vessel Type and Size

Module dimensions, weight, and origin/destination port characteristics influence the type and size of vessel(s) contracted for the project. Larger open deck vessels might be used for transporting very large modules. In the transportation report, several types of transportation vessels may be selected to define the design accelerations using the worst-case scenario. This might lead to a very conservative set of accelerations. Considering the range of module weights for each project, it is recommended to establish several design acceleration groups based on the size of vessel that will be used to transport the various module groups.

Towed barges could also be used for ocean transportation; however, barges typically have less capacity and move slower (6 knots) while self-propelled transport vessels are faster (12 knots) with higher capacity. Risk of losing cargo due to breakage of tow rope or tug propulsion failure is absent in the case of self-propelled transport vessels. Even when propulsion or steering fails on self-propelled vessels, the cargo will remain on the deck as opposed to cargo freely drifting in the ocean. Since self-propelled vessels are typically twin-screw vessels, propulsion failure related incidents are few.

Vessel speed is critical to circumvent events such as storms or hurricanes which could develop after tow was initiated and safe havens are either nonexistent or not quickly accessible.

6.3 Applicable Codes

During transportation, the authority having jurisdiction will be the Marine Warranty Surveyor (MWS).

The MWS is an independent third-party that performs technical reviews and approves the design and engineering drawings of modules for transportation purposes. In addition, the MWS performs a condition survey of vessels, conducts surveillance, and approves loadout and offload operations, and issues final approvals for sail away. Consultation with MWS at the early stages of projects is essential for alignment in design specifications and methodology. Project design methodologies regarding ocean transportation criteria and methodology including RISA 3D implementations for application of transportation forces shall be reviewed and approved by the MWS as early as possible on projects.



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As building codes do not cover marine transportation conditions, contract-specified specifications and standards published by Nobel Denton Marine Services such as DNV-ST-N001 typically apply. However, the design basis needs to be confirmed for each project. The ISO 19902 document for design of fixed steel offshore structures is another resource that could be used. The selected design basis must be approved by the Marine Warranty Surveyor (MWS), to receive their acceptance of the design approach.

6.4 Design Ocean Accelerations

The overall maximum ocean accelerations are based on comprehensive maximum response covering all headings and vessel types. Accelerations are typically provided in reports by the naval architect.

Some structural analysis programs such as STAAD Offshore can accept basic motion input, however, RISA-3D does not have this feature. When RISA-3D is used for analysis, it is required to calculate ocean forces resulting from design accelerations and apply them as nodal, point or distributed forces to the module structural model. For example, computed ocean acceleration forces due to dead loads will be applied at the member as a combination of nodal, point, and distributed forces. For this purpose, the WA-OCEAN spreadsheet is developed (refer to the Attachment 02 MS Excel file)

The WA-OCEAN spreadsheet takes all the acceleration values to calculate ocean forces on each single node of the RISA-3D model. Shown below is the input section of the WA-OCEAN file. Accelerations can be added in m/s^2 and rad/s^2 or changed to angular amplitudes and periods, as shown in Figure 2. Refer to Attachment 03 for the step-by-step procedure to design modules for ocean transportation.

Depending on project specifications dead load rotation might be included in design accelerations or it may be required to be added separately. For definition of dead load rotation, refer to Attachment 08. The spreadsheet provides an option to disable dead weight rotation, if required. Refer to RISA-3D Implementation for methodology of application of ocean acceleration to structural models.



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<u>(6) Input - Marine Accelerations:</u>		return to top	
<u>Translations:</u>			
Input using accelerations.	Change Translation Option		
	ampl (m)	period (s)	acceleration (m/s ²)
along longitudinal axis (surge)	5	10	1
along transverse axis (sway)	5	10	1
along vertical axis (heave)	5	10	1
<u>Rotations:</u>			
Input using accelerations.	Change Rotation Option		
	ampl (deg)	period (s)	acceleration (rad/s ²)
about longitudinal axis (roll)	18.38	8.97	1
about transverse axis (pitch)	4.66	10.63	1
about vertical axis (yaw)	3	10	1
Dead load rotation is enabled. (however rotation amplitude is also needed)	Change Dead Load Rotation Option		

Figure 2-Spreadsheet Inputs

6.5 Design Margins

Module weights and centers of gravity are subject to change during project execution. The recommended approach is to minimize engineering effort by implementing the following design margins for module transportation design:

- **Weight Envelope:** This envelope allows for uncertainty with the module dead weight at the time of analysis. Input load values in the Basic Load Cases typically represent the Estimated Weight (Gross Weight). Factors to modify the BLCs for the appropriate design weight (upper limit or lower limit) as provided in the module weight control report are by the module structural engineer and need to be incorporated into the designated component load combinations, described in Section 6.8.1 of this guideline. The module structural engineer needs to understand and provide appropriate factors.
- **Center of Gravity Envelope:** This envelope allows for uncertainty with the module center of gravity at the time of analysis. Envelope criteria, usually taken from the project module weight report, is entered into the WA-OCEAN spreadsheet. The spreadsheet then determines the envelope dimension to be applied as per the project design criteria and/or agreed to with the MWS as well as resulting additional forces.

For analysis, the CoG envelope shift is implemented as a combination of two effects. The first is a shift to align the RISA-3D CoG with the Weight Report CoG. The second is a shift to apply the CoG envelope. For more information, refer to Attachment 07.



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- **Positioning Envelope:** This envelope allows for uncertainty with module positioning on the vessel deck at the time of analysis. Positioning uncertainty criteria is entered into the WA-OCEAN spreadsheet. The spreadsheet then evaluates the positioning criteria and determines associated dimensions to the center of rotation. Unless the position of the module on the vessel is confirmed at the time of design, modules should be checked at locations on the vessel resulting in maximum design forces for the module and grillage and sea fastening.

Even if module positions on vessels are known at the time of the structural analysis, modules should still be checked at other possible locations on the vessel as changes to vessel voyages can affect stowage plans.

- **Utilization Ratios:** the ratio of calculated load divided by calculated capacity. Codes and specifications permit this value to be a maximum of 1.0, however a lower value may be appropriate to deal with design uncertainty at the time of analysis. Discuss with your lead engineer about the maximum utilization ratios acceptable for transportation analysis.

6.6 Ocean Wind

One of the recommended practices with a procedure on ocean wind calculation is DNV-ST-N001 Marine Operations and Marine Warranty, Section 3.4.6.

It should be noted that the “ground” elevation used in the ocean transportation wind load calculation is at the water surface, not the base of the structure.

6.7 Vessel Longitudinal Bending (Hog and Sag)

Hog and Sag are the longitudinal bending of the deck causing both vertical and horizontal deflections resulting from ship flexing due to interaction with waves. Because they are not uniform and deflect the ship along its length due to the length of module transportation vessels relative to wave spacing, these deflections impart stresses on the module, similar to how a settling foundation affects a building structure. For modules much shorter than the length of the ship (shorter than 1/3 of the length of the ship per Section 5.6.11 of the DNVGL), sag and hog may be ignored if indicated by the Naval Architect and approved by the MWS.

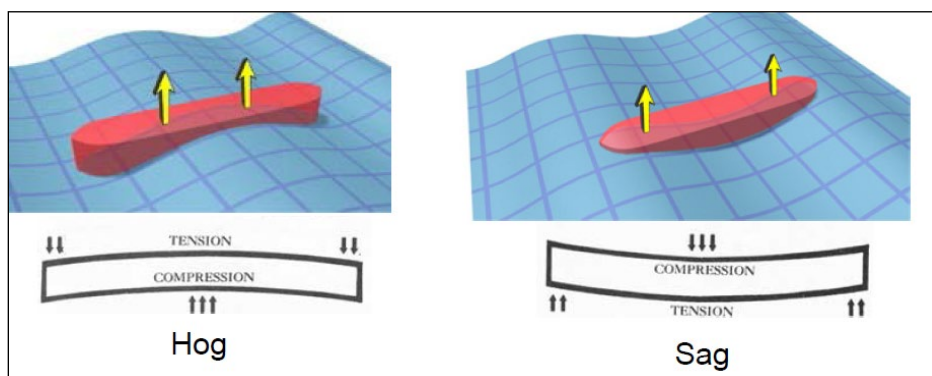


Figure 3-Hog and Sag

To calculate the deflection of the vessel under each support point of the module, a vessel deflection curve is required. The deflection curve could be computed by the Naval Architect or



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provided in the project specifications. It is usually simplified by a 180-degree sinusoidal curve. Consultation and agreement with the MWS is necessary before using the simplified curve for sag and hog analysis. Often unstated, maximum sag and hog values apply halfway between the vessel deck and keel and will result in a horizontal component at the deck level. The hull girder deflection provided is applicable at the neutral axis of the hull structure. The neutral axis will be located close to mid depth. The plating thicknesses of the deck and the bottom will be the main drivers in determining its location. Typically, a set of discrete values is converted to equation form, using a sinusoidal curve. Refer to Attachment 09 for details of the calculation.

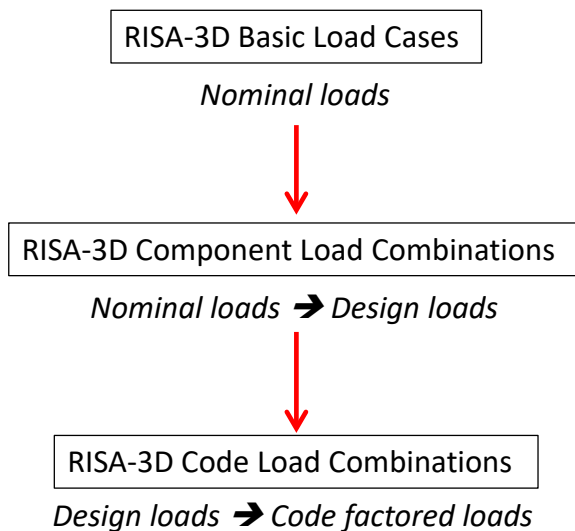
6.8 Structural Model Implementation

Ocean inertia forces on transported modules may be applied to structural models with implementation of the center of gravity (COG) shift/tolerance. This section provides an approach to calculate and input ocean transportation loading into a RISA-3D model, which is not applicable to other structural analysis programs including STAAD Pro Offshore (recommended) or SACS. This approach is an efficiency-driven workaround until a comparable capability is added into the RISA-3D program. The general approach described in this section includes:

6.8.1 Basic Load Cases and Load Combinations

Project basic load cases and load combinations (strength-level and service-level load combinations) should be set up in a format that can be copied directly into RISA-3D.

- Basic load cases (BLCs) need to be included for input of acceleration nominal loads.
- Component load combinations (CLCs) need to be included to convert nominal loads into design values.
- Code load combinations (LCs) need to be included to implement code-specified load factors.





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Figure 4-Loading Scheme

a. BLCs for Dead Weight

Five BLCs are created for transportation dead weight and locations of CoG shifts:

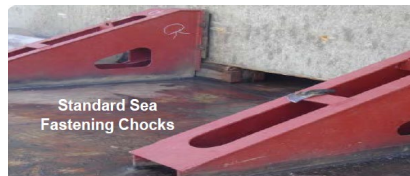
Basic Load Cases X		
	BLC Description	Category
100	***TRANSPORTATION DEAD WEIGHT***	None
101	Temporary Steel	None
102	Grillage Steel	None
103	Seafastening Steel	None
104	Stowage Load	None
105	CG shift: locations	None

Modeled transportation member weights should already be included with the RISA-3D self-weight factor. The distinction and applicable usage of the below-listed BLCs is up to the Project Lead's direction but is generally in accordance with the following.

Temporary Steel is internal to the module that is installed at the module fabrication yard and removed after final setting of the module at the project site.

Grillage Steel is steel below the module designed to fully support modules during ocean transportation, enabling the transportation loads to be safely distributed to the deck and subframes of the vessel. Grillage is installed prior to loadout and removed after setting. Grillage should accommodate loading and unloading strategies and constraints including height and stroke of SPMTs.

Seafastening Steel connects the module to the marine vessel deck to secure modules to the vessel deck, and is installed just after loadout, and removed just prior to offload.





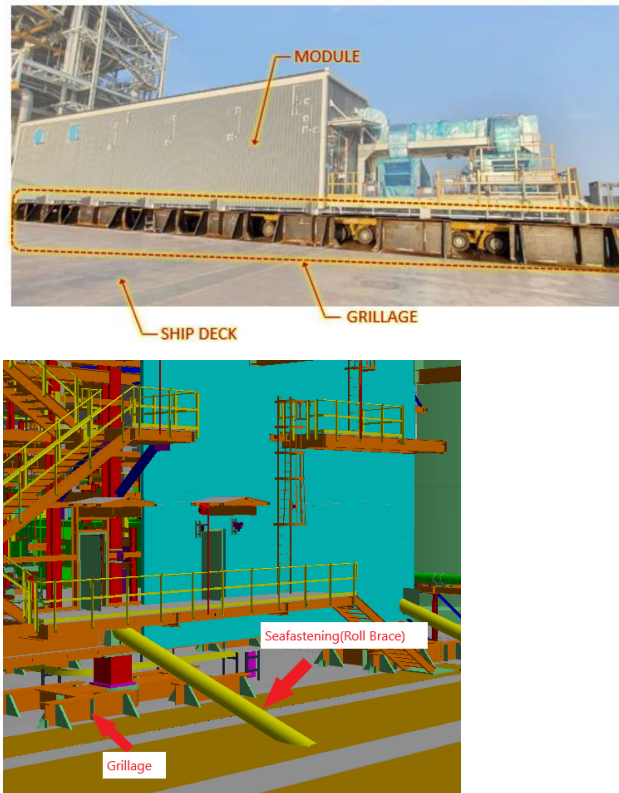
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Figure 5-Grillage and Sea fastening

Stowage Load is for loose items transported within a module, added prior to loadout, and removed after setting. Typically, a main deck uniform floor load is applied with a value provided in the project structural criteria specification.

CoG shift: locations are used to define the nodes that will be used to apply center of gravity shift forces as described in Section 6.8.6.

b. BLCs for Wind Loads

Four BLCs are created for wind during marine transport.

If both the wind forces and the structure are symmetrical (i.e. $+X = -X$ or $+Z = -Z$), then only two of the BLCs would be needed with direction factors applied to the component load combination.

If wind loads are to be ratioed from site wind BLCs, then refer to those BLCs and apply factors for the difference in wind pressure. The height above finished grade and above the ocean surface should be considered.



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c. BLCs for Hog and Sag

Two BLCs are created for enforced displacements due to deck flexing. For hog and sag calculation and application in RISA-3D, refer to Section 6.7 and Attachment 09.

Basic Load Cases X		
	BLC Description	Category
106		None
107	****MARINE TRANSPORT****	None
108	***Marine Wind Load***	None
109	+WMT: Wind Transverse +X directi	None
110	-WMT: Marine Wind Transverse -X	None
111	+WML: Marine Wind Longitudinal +	None
112	-WML: Marine Wind Longitudinal -	None
113	---	None
114	***Deck Deflection***	None
115	Vessel Hog (Bh)	None
116	Vessel Sag (Bs)	None

d. BLCs for Nominal Marine Accelerations

Four sets of basic load cases are created for nominal marine accelerations, one set for each envelope corner. The envelope corners are related to the four positions of module on the ship which are fore/port, fore/starboard, aft/port, and aft/starboard. "Fore" refers to front (bow) of the ship, "Aft" refers to rear or stern of the ship, "Port" refers to the left side of the ship facing the bow (front) and "Stbd" refers to starboard which is right of the ship facing the bow (front).

Each set consists of three translation BLCs and three rotation BLCs as described below. The designations ("NTX1:", "NRX1:" etc.) are used by WA-OCEAN macros for search purposes. Refer to Attachment 04 for recommended axis orientations.

Set 1 is for position FP which is fore/port or when facing the bow, forward/left.

Basic Load Cases X		
	BLC Description	Category
124	***module position FP***	None
125	(Nominal Translational Accel)	None
126	NTX1: along X axis	None
127	NTY1: along Y axis	None
128	NTZ1: along Z axis	None
129	---	None
130	(Nominal Rotational Accel)	None
131	NRX1: about X axis	None
132	NRX1: about Y axis	None
133	NRX1: about Z axis	None



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Set 2 is for position FS which is fore/starboard or when facing the bow, forward/right.

Basic Load Cases X		
	BLC Description	Category
135	(module position FS)	None
136	(Nominal Translational Accel)	None
137	NTX2: along X axis	None
138	NTY2: along Y axis	None
139	NTZ2: along Z axis	None
140	---	None
141	(Nominal Rotational Accel)	None
142	NRX2: about X axis	None
143	NRX2: about Y axis	None
144	NRZ2: about Z axis	None

Set 3 is for position AP which is aft/port or when facing the bow, behind/left.

Basic Load Cases X		
	BLC Description	Category
146	***module position AP***	None
147	(Nominal Translational Accel)	None
148	NTX3: along X axis	None
149	NTY3: along Y axis	None
150	NTZ3: along Z axis	None
151	---	None
152	(Nominal Rotational Accel)	None
153	NRX3: about X axis	None
154	NRX3: about Y axis	None
155	NRZ3: about Z axis	None

Set 4 is for position AS which is aft/starboard or when facing the bow, behind/right.

Basic Load Cases X		
	BLC Description	Category
157	(module position AS)	None
158	(Nominal Translational Accel)	None
159	NTX4: along X axis	None
160	NTY4: along Y axis	None
161	NTZ4: along Z axis	None
162	---	None
163	(Nominal Rotational Accel)	None
164	NRX4: about X axis	None
165	NRX4: about Y axis	None
166	NRZ4: about Z axis	None



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e. BLCs for CoG Shift

Four basic load cases are created for nominal CoG shifts. The designations “FPU:”, “FSU:”, “APU:” and “ASU:” are used by WA-OCEAN macros for search purposes which represent fore/port/up, fore/stbd/up, aft/port/up, and aft/stbd/up, respectively. Instructions are provided within the spreadsheet. CoG shifts are considered to be upwards not downwards for analysis purposes as upwards shifts would increase the structure loads not downward shifts.

Basic Load Cases X		
	BLC Description	Category
118	***CG Shift Loads***	None
119	FPU:fore/port/up	None
120	FSU:fore/stbd/up	None
121	APU:aft/port/up	None
122	ASU:aft/stbd/up	None

WA-OCEAN spreadsheet should be used for the Basic load cases (CoG shift and position shift and marine nominal accelerations). Instructions are provided within the spreadsheet. Refer to Attachment 02 for step-by-step instruction.

f. Calculation and Input of Component Load Combinations

All component load combinations must convert loading from the RISA-3D coordinate system used in the BLCs into the transportation LTV system used in the code load combinations. Derivation of LTV coordinate system is described in Attachment 04. The WA-OCEAN spreadsheet assists with the inertia-related parts of this conversion.

Using different coordinate systems brings a potential for errors in analysis. If alignment of coordinate systems is not possible, diligent checking is necessary after coordinate conversion is completed.

Component Load Combinations for Dead Weight

Two main component load combinations are defined for total dead weight: “Dship” and “Iship”. Dship represents all module design dead weight during ocean transportation including all pipe dead loads. “Iship” is required if pipe stress transportation forces are applied separately from the inertia calculation. In this case dead weight of pipes (that marine inertia forces are provided on stress ISOs), shall be excluded from the total dead weight input to WA-OCEAN spreadsheet to avoid double counting. Therefore, “Iship” equals “Dship” minus pipe empty weight of pipes (which ocean transportation forces are applied separately based on Stress ISOs). “Iship” are computed and used internally by the WA-OCEAN macro.



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Load Combinations X				
Combinations Design				
LC Generator RSA Scaling Factor				
	Description	Solve	PDelta	SRSS
1300	{ Transportation Loading, DNV-ST-N001 (2016-06) }	☐		
1301	(Ocean Load Components)	☐		
1302	Dship: unshifted transportation dead weight	☐		
1303	Iship: unshifted transportation inertia weight	☐		

“Dship” needs to have four new component load combinations to compute the CG shift for each 4 CG envelope corners. Therefore, “DshipFP”, “DshipFS”, “DshipAP”, and “DshipAS” are defined to represent additional dead loads added to shift RISA-3D CoG to each 4 CoG envelope corners. References and factors for these CLCs will be calculated by the WA-OCEAN spreadsheet.

Load Combinations X				
Combinations Design				
LC Generator RSA Scaling Factor				
	Description	Solve	PDelta	SRSS
1312		☐		
1313	(Dship, with CG shift)	☐		
1314	DshipFPU: fore/port/up	☐		
1315	DshipFSU: fore/stbd/up	☐		
1316	DshipAPU: aft/port/up	☐		
1317	DshipASU: aft/stbd/up	☐		

The designations (“Dship:”, “Iship”, “DshipFPU”, “DshipFSU”, etc.) are used by WA-OCEAN macros for search purposes.

Note that CoG shifted inertia forces (“Iship”) are computed and used internally by the WA-OCEAN macro.

Inertia CLCs are specific to each corner of the positioning envelope (FP, FS, AP, AS). For the needs of specific projects, the following generic CLCs may need to be modified.

Component Load Combinations for Ocean Inertia Forces

A total of 40 CLCs are defined for ocean inertia forces as follows. For head sea conditions,



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Row	Label (max 79 chars)	Solve	Pdelta	SRSS
1194	(Head Sea, position FP)			
1195	H1: +(surge+pitch) +(heave)			
1196	H2: +(surge+pitch) -(heave)			
1197				
1198	(Head Sea, position FS)			
1199	H3: +(surge+pitch) +(heave)			
1200	H4: +(surge+pitch) -(heave)			
1201				
1202	(Head Sea, position AP)			
1203	H5: -(surge+pitch) +(heave)			
1204	H6: -(surge+pitch) -(heave)			
1205				
1206	(Head Sea, position AS)			
1207	H7: -(surge+pitch) +(heave)			
1208	H8: -(surge+pitch) -(heave)			

For beam sea conditions,

Row	Label (max 79 chars)	Solve	Pdelta	SRSS
1210	(Beam Sea, position FP)			
1211	B1: +yaw +(sway-roll) +heave			
1212	B2: +yaw +(sway-roll) -heave			
1213	B3: -yaw +(sway-roll) +heave			
1214	B4: -yaw +(sway-roll) -heave			
1215				
1216	(Beam Sea, position FS)			
1217	B5: +yaw +(sway-roll) +heave			
1218	B6: +yaw +(sway-roll) -heave			
1219	B7: -yaw +(sway-roll) +heave			
1220	B8: -yaw +(sway-roll) -heave			
1221				
1222	(Beam Sea, position AP)			
1223	B9: +yaw -(sway-roll) +heave			
1224	B10: +yaw -(sway-roll) -heave			
1225	B11: -yaw -(sway-roll) +heave			
1226	B12: -yaw -(sway-roll) -heave			
1227				
1228	(Beam Sea, position AS)			
1229	B13: +yaw -(sway-roll) +heave			
1230	B14: +yaw -(sway-roll) -heave			
1231	B15: -yaw -(sway-roll) +heave			
1232	B16: -yaw -(sway-roll) -heave			

For quartering sea conditions,



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Row	Label (max 79 chars)	Solve	Pdelta	SRSS
1234	(Quartering Sea, position FP)			
1235	Q1: +yaw +0.6(surge+pitch) +0.8(sway-roll) +(heave)			
1236	Q2: +yaw +0.6(surge+pitch) +0.8(sway-roll) -(heave)			
1237	Q3: -yaw +0.6(surge+pitch) +0.8(sway-roll) +(heave)			
1238	Q4: -yaw +0.6(surge+pitch) +0.8(sway-roll) -(heave)			
1239				
1240	(Quartering Sea, position FS)			
1241	Q5: +yaw +0.6(surge+pitch) -0.8(sway-roll) +(heave)			
1242	Q6: +yaw +0.6(surge+pitch) -0.8(sway-roll) -(heave)			
1243	Q7: -yaw +0.6(surge+pitch) -0.8(sway-roll) +(heave)			
1244	Q8: -yaw +0.6(surge+pitch) -0.8(sway-roll) -(heave)			
1245				
1246	(Quartering Sea, position AP)			
1247	Q9: +yaw -0.6(surge+pitch) +0.8(sway-roll) +(heave)			
1248	Q10: +yaw -0.6(surge+pitch) +0.8(sway-roll) -(heave)			
1249	Q11: -yaw -0.6(surge+pitch) +0.8(sway-roll) +(heave)			
1250	Q12: -yaw -0.6(surge+pitch) +0.8(sway-roll) -(heave)			
1251				
1252	(Quartering Sea, position AS)			
1253	Q13: +yaw -0.6(surge+pitch) -0.8(sway-roll) +(heave)			
1254	Q14: +yaw -0.6(surge+pitch) -0.8(sway-roll) -(heave)			
1255	Q15: -yaw -0.6(surge+pitch) -0.8(sway-roll) +(heave)			
1256	Q16: -yaw -0.6(surge+pitch) -0.8(sway-roll) -(heave)			

g. Code Load Combinations

Code load combinations should contain references only to the BLCs described above. Thus, load combination references and load factors are to be included in the project standard and copied by each module engineer into RISA-3D.

All code load combinations use the transportation LTV coordinate system. Projects may need to apply modifications to the generic list.

The code load combinations described in Attachment 06 are based on DNVGL-ST-N001, Marine Operations and Marine Warranty. DNVGL section 5.6.15.1 permits the combination of peak values.

6.8.2 Boundary Conditions

Because module analysis involves the use of multiple design situations (SPMT transportation, marine transportation, weighing, crane lifts, etc.) which involve unique sets of boundary conditions, multiple analysis models are traditionally used. The drawback to this approach is that any design development change must be made to every analysis model being used. A method has been developed to use a single RISA-3D model for multiple design situations and includes a means to switch between boundary conditions with minimal effort. This single model approach uses,

- Enabled and disabled members to distinguish between members which are to be included or excluded from the desired design situation. For example, SPMT members are disabled for



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marine transportation. Temporary members are enabled only for transportation situations. Note that boundary conditions connected only to disabled members will have no effect on the analysis. When switching between design situations, there is no need to add or remove any members.

- Tethered nodes (formerly master-slave nodes) are used to establish different boundary conditions for a single member. An example of this is presented in Figure 7. Note that dimensional differences between tethered nodes are, by definition, not considered by RISA-3D. Thus, the column base shown in Figure 7 will behave as a single column with no offset, and with a variable boundary condition at its base. When switching between design conditions, there is no need to alter any tethered node settings.
- A single set of boundary conditions is established for all design situations. Enabled and disabled members will determine which boundary conditions are in use. When switching between design situations, there is no need to alter any boundary conditions.
- Material labels are used to categorize members by design situation. This facilitates the switching process. For example, a permanent grade 50 member would have material "Perm-50". A member only intended for marine transportation would use material "Sea-50". An SPMT spine beam would use material "SPMT-spine". When switching between design situations, there is no need to alter any material definitions.

To switch from one design situation to another, the process would be to (1) enable all members, (2) unselect all members, (3) using material labels, select members that are not needed for the next design situation, (3) disable selected members, (4) manually update the list of load combinations to be analyzed.

Module stub column support arrangements can vary as shown in Figure 6. Variations can also include modules without legs or heavy modules that are supported directly from the base frame. The modeling that is used will require appropriate attention based on the concepts discussed in the following.





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Figure 6- Stub Columns During Marine Transport

Modeling consists of a detail at the base of each stub column to represent the different analysis conditions. For marine transportation, members associated with site and SPMT conditions need to be disabled as indicated in Figure 7b. This configuration can be modeled once, and then copied to each column location as needed.

The “site” member segment should be modeled directly below the stub column (Figure 7b). For dynamic analysis conversions to SAP2000, this would facilitate the transition. Note that the offset marine and SPMT member segments will have a slight effect on the CoG calculation, therefore, it is recommended to minimize the offset when modeling member segments.

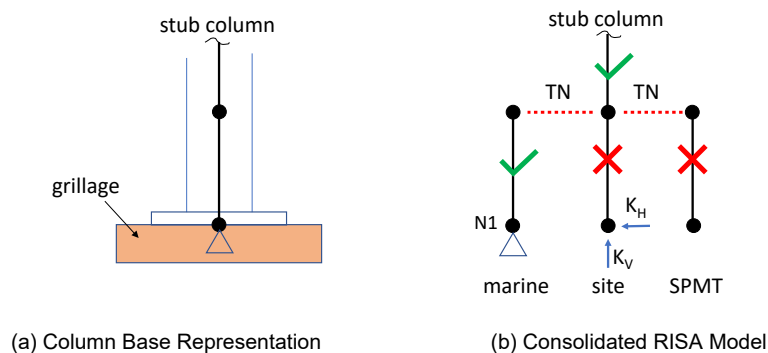


Figure 7- Modeling of Stub Column on Vessel Deck

In Figure 7, checkmarks indicate enabled members and “TN” indicates a tethered node (formerly master-slave) boundary condition. All three sub column segments have the same material and shape properties. Node “N1” in Figure 7b is where hog/sag displacements are specified.

6.8.3 Modeling Equipment

With the center of gravity to be compared to the value provided in the Weight Report, it is advisable to model large equipment as accurately as reasonable. For conventional stick-built structures, equipment loads are typically provided as applied forces at the equipment base. For accurate marine load calculations, large equipment items should be modeled using rigid weightless elements to position equipment CoGs at the correct elevation.



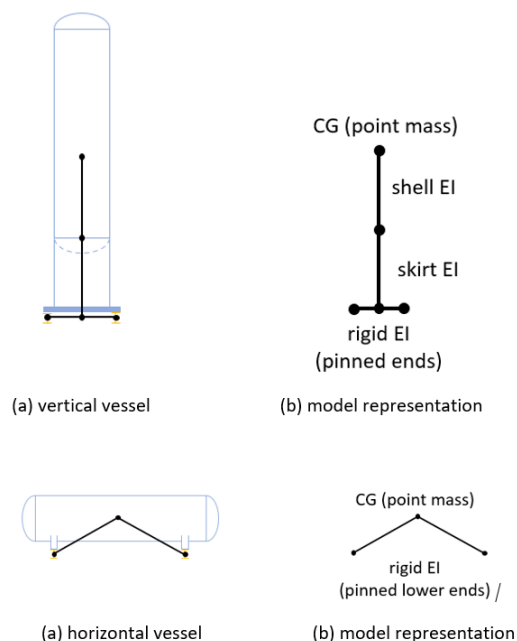
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Figure 8- Modeling of Equipment

Rigid links are used to model all or parts of mechanical equipment. RISA-3D implements rigid links by using weightless “rigid” members (frame members with very high section properties). Care must be taken with short rigid members to avoid numerical issues. RISA-3D will report such issues as a warning (i.e., “Sum of Y reaction is not equal to the sum of the loads (LC ##)!”). In addition, if environmental thermal load case is used in RISA-3D for in-place condition, rigid links might report unrealistic member forces.

For tall vertical vessels transported on modules, it may become desirable to add braces near the top of the vessel. Always work this issue with the Mechanical group to avoid unintended consequences.

6.8.4 Modeling Area and Surface Loads

The WA-OCEAN spreadsheet does not handle area loads or surface loads. It is preferred to avoid the use of these load types. However, if used, these load types will need to be converted to distributed loads. The following sequence can be applied.

Step 1: Find the load combination which will be used for the WA-OCEAN dead load calculation (“Dship”) and record all referenced basic load cases and basic load categories. In this example, these are categories “DL” and “OL1”.



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Load Combinations								
Combinations		Design						
LC Generator		RSA Scaling Factor		Solve Current LC		Solve Batch + Envelope		
	Description	Solve	P-Delta	SRSS	BLC	Factor	BLC	Factor
552	(Ocean Load Components)	☐						
553	Dship: unshifted transportation dead weight	☐			DL	1	OL1	1
554	Lship: unshifted transportation inertia weight	☐			DL	1	OL1	1

Step 2: For each referenced basic load case, identify if any area/surface loads are applied. In this example, an area load is shown for “DL”.

Basic Load Cases									
	BLC Description	Category	X Gravity	Y Gravity	Z Gravity	Nodal	Point	Distributed	Area(Member)
1	** STRUCT AND ARCH DEAD WEIGHT *	None							
2	Main Frame Steel (Ds)	DL		-1.08					
3	Intermediate Steel (Ds)	DL				80		94	
4	Grating (Ds)	DL				2	3	120	1
5	Floor Plate (Ds)	DL						86	
6	Handrail (Ds)	DL				2		4	
7	Stair/Ladder (Ds)	DL							

Step 3: Have RISA-3D solve the designated mass load combination. Then search the list of BLCs for a transient BLC matching the original BLC row number. In the example shown, a transient load is shown for BLC 4.

Basic Load Cases									
	BLC Description	Category	X Gravity	Y Gravity	Z Gravity	Nodal	Point	Distributed	Area
142	---	None							
143	BLC 4 Transient Area Loads							2398	
---	---								

Step 4: Click on the transient load row at the distributed loads. Select the list of distributed loads and copy into an empty BLC and set the category to DL.

In the original BLC row, delete the area load.

6.8.5 Modeling Hog and Sag Displacement

Hog and sag displacements are applied in RISA-3D basic load cases. The magnitude of these displacements is described in section 6.7. Due consideration should be given to horizontal components of hog/sag displacements. Refer to Attachment 09 for hog and sag detail calculation.

Note that displacements can be applied at a boundary condition (like N1 in Figure 7.b), but a theoretical contradiction is created between the boundary condition and the enforced



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displacement. RISA-3D resolves this situation by using the enforced displacement and ignoring the boundary condition.

- Imposing a displacement at any node implies the application of an external force. The value of this force (the reaction in this case) is not reported but can be determined from member forces in connected members.
- The boundary condition cannot be removed if it is necessary to keep the model theoretically stable.
- Forces due to imposed displacements are determined from member stiffnesses within the module. When the AISC Direct Analysis Method is used, with associated reduced stiffness, then a 20% allowance is suggested to compute correct force magnitudes.

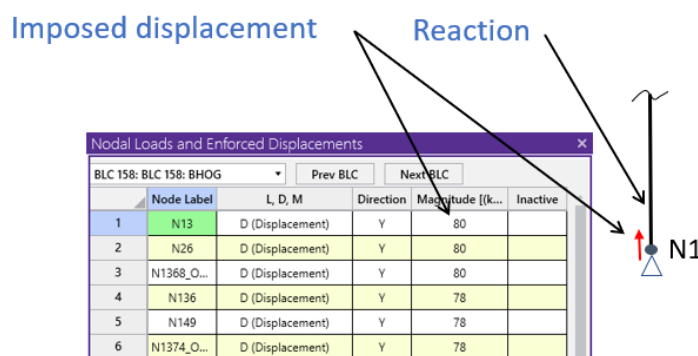


Figure 9- Imposed Displacements

Hog and sag values are provided by the Naval Architect or from the Project Specifications. Often unstated, these values apply halfway between the vessel deck and keel and will result in a horizontal component at the deck level. Typically, a set of discrete values is converted to equation form, using either a parabolic or sinusoidal approach. Refer to Attachment 09 for details of the calculation.

6.8.6 Modeling for Center of Gravity Shifts

CoG shifts to apply a CoG envelope are part of the design margins described in section 6.5.

Because actual local CoG shift details cannot be predicted, CoG shift loading is considered a global effect. There is also an observed need to avoid CoG shift forces on modeled tertiary members. Thus, a list of user-specified nodes is needed to identify the main member nodes that will be used by WA-OCEAN. User-specified nodes need to be located on all sides of the anticipated center of gravity and need to account for connections of all main beams framing into columns forming the structure's primary load paths for vertical loads to the base of the module.

A RISA-3D selection state could have been used, but the approach is not practical for visual confirmation. So, the WA-OCEAN spreadsheet uses a user-selected basic load case ("CoG Shift locations"). The spreadsheet uses the selected BLC only to retrieve a list of nodal force nodes. Point loads, distributed loads, area loads, and surface loads are neglected. The magnitudes of applied nodal forces under CoG shift BLC will have no impact on the analysis.



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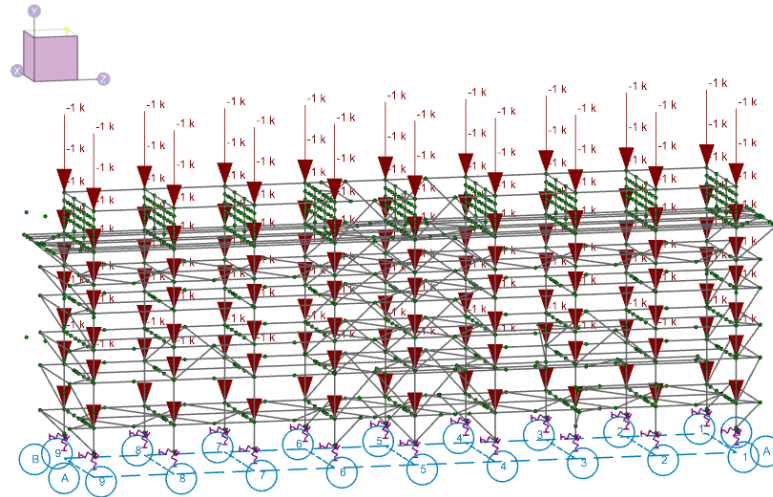


Figure 10- Sample Image of CoG shift load locations for BLC 105 “CoG Shift locations”

6.9 Other Design Considerations

6.9.1 Fatigue

Fatigue is typically significant for offshore platform modules with their entire service life exposed to wave loading. Fatigue design for offshore facilities is an established and necessary component of the overall design process. Fatigue loading for onshore modules, during ocean transport, is often regarded lightly due to the relatively short duration for modules to experience repeated cyclic loading. However, a simple and conservative examination has usually been sufficient. If the analysis results indicate that fatigue needs to be accounted for, a full spectral fatigue analysis should be performed. Design Considerations such as the "total cumulative damage" to be able to eliminate fatigue might be an option to opt out fatigue design from the analysis. Fatigue design philosophy should be reviewed and approved by MWS.

Onshore structural specifications which cover fatigue are ANSI/AISC 360-16 (Appendix 3), DNV-ST-N001, Section 5.9.4, and AWS D1.1 (Section 2, part C).

ANSI/AISC 360-16, Appendix 3, Section 3.1 covers fatigue design for steel members and indicates no evaluation of fatigue resistance is required if the number of loading cycles is less than 20,000, or if the live load stress range, associated with predominant waves is less than threshold stress range FTH (Refer to ANSI/AISC 360-16, Appendix 3, Table A-3-1).

6.9.2 Load-on/Load-out (Roll-on/Roll-out – RO-RO)

Using SPMTs (Self Propelled Modular Transporter) is the most common and most preferred method of transferring a very large module from the place of fabrication to a seagoing vessel and from the vessel to its final position. Refer to Fluor Guideline 000.215.0766 for design consideration for elevated and embedded base frames.

SPMT transportation occurs before and after ocean transportation. SPMT analysis should start at an early stage of the project, even if the module land transportation contractor submittals are not



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available and expected to be provided at later stage of the project. Alignment between the SPMT design requirements (locations, etc.) is preferable to avoid multiple design cases from an analysis perspective.

In addition, SPMT transportation is not limited to RO-RO as SPMTs are also used to move modules to positions for lift-on/lift-off operations, described in Section 6.9.3.

During land transportation on SPMT, modules are subjected to internal forces due to the SPMT's braking forces, road slope/camber, wind and snow loads. The following are some key points required for SPMT analysis, however detailed procedure is out of scope of this guideline.

Transportation Arrangement (TA) drawing shall be provided by module land transportation contractor for each module. Verify and document the following design criteria with the contractor:

- Individual trailers – Size, weight, capacity, manufacturer, model
- Power Pack units – Size, weight, CoG, manufacturer, model, maximum number of driven trailers
- Spine beam – dimensions, moment of inertia, allowable bending, allowable shear
- Cylinders – maximum & minimum possible stroke, permissible relative deflection
- At rest & transport elevations, and vertical clearances associated with each SPMT move
- Minimum horizontal clearances
- Hydraulic grouping

Coordination is required to clarify the clearance verification responsibility split between Fluor and the heavy haul contractor(s). **The height of the grillage and the elevation of foundations are two key considerations when verifying SPMT clearances.**

Different configuration requirements between module yard and site heavy haul contractors. A center of gravity shift would be caused by changes in the distribution of weight within a module. At the time of analysis, the details causing a shift would be unknown. Thus, a CoG shift is defined to account for unknowns. Design consideration for CoG shifts including CoG shift envelope shall be communicated to module land transportation contractor at early stage of the project.

Ship loose items shall be defined during design stage and appropriate loading shall be applied to the structural model to account for ship loose items during transportation. Refer to Fluor Guideline 000.215.0766 for some discussion on ship loose items.

6.9.3 Lift On/Lift Off (Lo-Lo)

A type of installation using a crane lift to load modules to and from a vessel. Lifting analysis needs to carry out to verify whether the structure has adequate capacity to withstand the loading during the lift condition, without imposing undue and unanticipated forces on assembled equipment, piping and components and to determine the lift forces acting at the lift locations for design of auxiliary members that form part of the lifting system. Generally, CSA team is involved in verification of Global structural integrity of modules per coefficients given in Section 16 of DNVGL-ST-N001 and provide the reactions / forces to logistics for further action.



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Figure 11- Pipe rack, Coker Structure, Module and Very Large Module lifting

(Photo Courtesy: 000.215.0766, www.ale-heavylift.com, DSME Shipyard Okpo)

The flow chart in Figure 12 shows the approach & stakeholders involved during complete lifting / rigging operation and where & how structural analysis of modules fits in entire process. Blue Box mentioned in below chart indicates the activity / task performed by CSA design engineer considering factors mentioned in Green Boxes.

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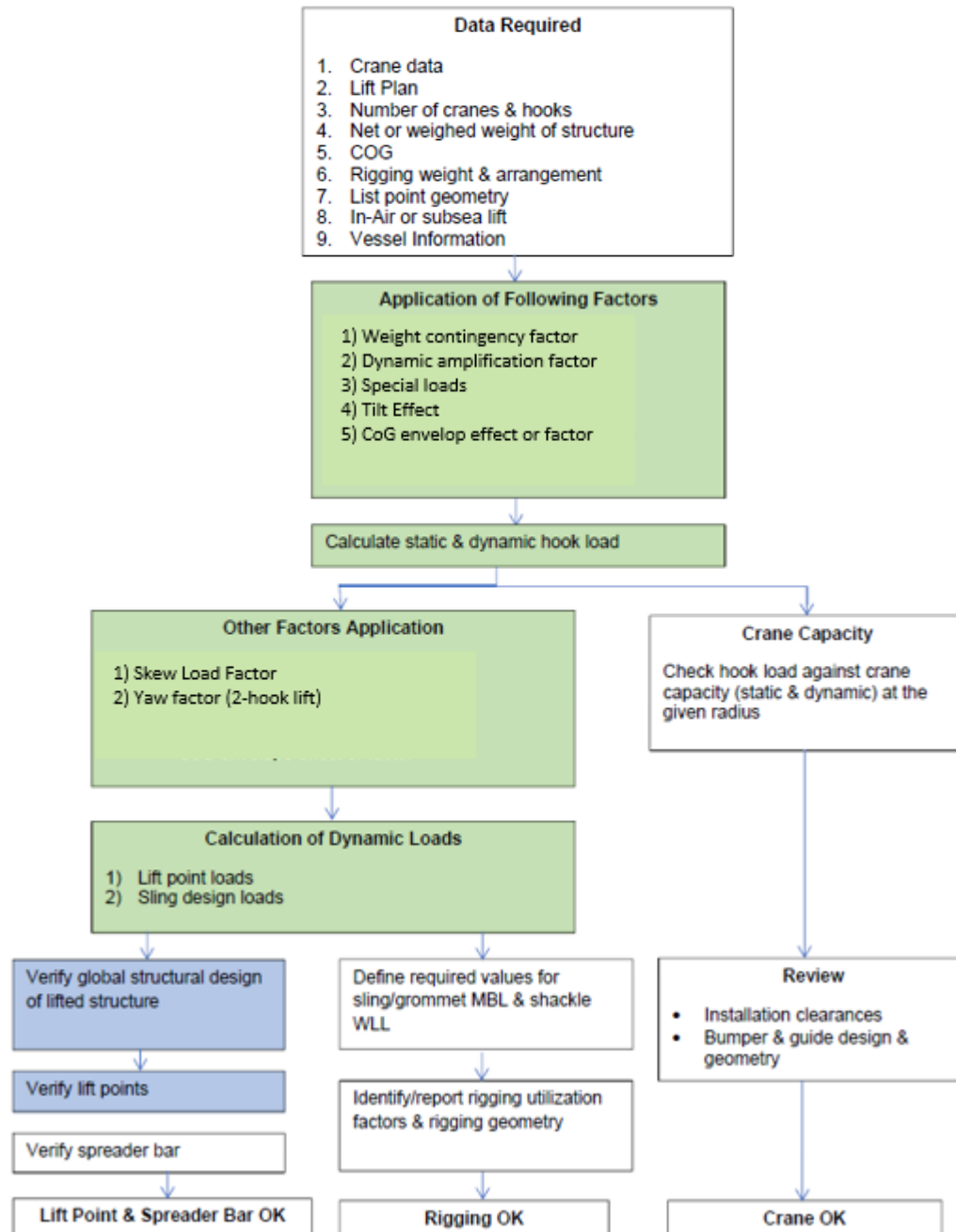


Figure 12- Lift calculation Flowchart (Source: DNVGL-ST-N001, Figure 16-1)



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Per above flow chart modules are designed for dead load, plus vertical impact load. Engineers need to ensure to take appropriate factors as per specific project requirements or design basis, as these forces also govern the design and safety of lifting attachments. Vertical impact due to lifting and lowering of modules by crane is included in module design loads.

Following is the table taken from one of the projects as sample to show the sample values of these factors. This may vary depending on the lifting conditions and project requirements.

Lift Factor		Factor
Weight Contingency Factor, f_1 <i>DNVGL-ST-N001 Section 16.2.2</i>	Weight of module derived by calculation	1.10
	Module accurately weighed	1.05
Center of Gravity Factor, f_2 <i>DNVGL-ST-N001 Section 5.6.2.3</i> (Use of CoG envelope is highly recommended.) Note: it is required to establish a linear behavior between CoG shift and corresponding increase in stress, support reactions, etc. The "CoG envelope not used" factor can only be used if the direct linear relationship between the CoG envelope and the factor is established. This is only true for very rigid structures.	CoG envelope not used	1.10
	CoG envelope used	1.05
Dynamic Amplification Factor, f_3 where land-based crane is used. <i>DNVGL-ST-N001 Section 16.2.5</i>	Gross Lifted Weight < 100 tonnes	1.10
	100 tonnes < Gross Lifted Weight < 1000 tonnes	1.05
	1000 tonnes < Gross Lifted Weight < 10000 tonnes	1.03
Dynamic Amplification Factor, f_3 where vessel's crane is used. <i>DNVGL-ST-N001 Section 16.2.5</i>	Gross Lifted Weight < 100 tonnes	$1.07 + 0.05 \times (100/\text{Gross Lifted Weight})^{0.5}$
	100 tonnes < Gross Lifted Weight < 300 tonnes	1.12
	100 tonnes < Gross Lifted Weight < 1000 tonnes	1.10



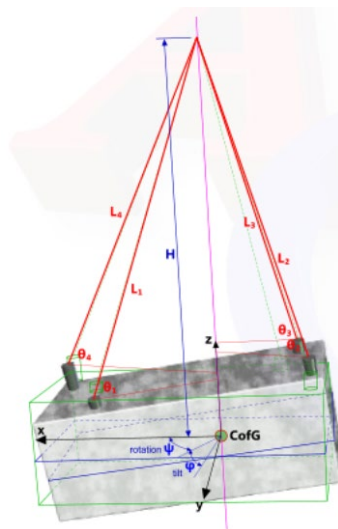
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Lift Factor		Factor
	1000 tonnes < Gross Lifted Weight < 2500 tonnes	1.08
	2500 tonnes < Gross Lifted Weight < 10000 tonnes	1.05
Skew Load Factor, f_4 (To account additional loading caused by fabrication tolerances and other uncertainties with respect to asymmetry and associated force distribution in the rigging arrangement) <i>DNVGL-ST-N001 Section 16.2.6</i>	Indeterminate lifts with 4-sling lifts without spreader bars, if the conditions provided in DNVGL-ST-N001 Section 16.2.6.8 are met. Note: the factor shall be applied to induce a racking effect to each diagonally opposite pair of lift points.	1.25
	Factor for basic lift design, and with lift system incorporating spreader bars using matched pair of slings, if the conditions provided in DNVGL-ST-N001 Section 16.2.6.6 for single hook lift or 16.2.6.7 for a tandem lift are met.	1.10
	Statically determinate lift system (for example a stiff skid, where the rigidity of the system does not allow for any racking)	1.00
Yaw Factor, f_5 (To account for increased sling loading due to rotation of the module about vertical axis) <i>DNVGL-ST-N001 Section 16.2.4</i>	For 2-hook lifts, to account for increased sling loading caused by rotation of the object about a vertical axis. Yaw factor should be applied	1.05
Minimum Sling Angle (to the horizontal)		45°
Member Consequence Factor (MCF), f_6 <i>DNVGL-ST-N001 Section 16.8.3</i>	Lift points and lift point attachments to structure	1.30
	Members directly supporting or framing into lift points	1.15
	Other members	1.00

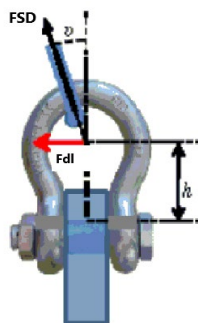
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Table 1: Lifting Factors (extracted from DNV-ST-N001: For Reference Only)

Tilt Effect (The angle of tilt (ϕ)): A tilt effect shall be calculated to account for increased sling loading caused CoG position and/or horizontal imposed loads for a single hook lift (*DNVGL-ST-N001 Section 16.2.3*) or by rotation of the object about a horizontal axis and the effect of out of plumb hoist lines for a tandem lift (*DNVGL-ST-N001 Section 16.2.4*). The tilt effect shall be 2 degrees for a single hook lift under normal circumstances. The tilt effect shall be 3 degrees or hook elevation difference of +/- 1.0m (whichever is more onerous) for a tandem lift using two cranes on one vessel, and 5 degrees or hook elevation difference of +/- 1m (whichever is more onerous) for a tandem lift using two separate vessels. The angle of tilt (ϕ) is shown in the figure below.


Figure 13- Angle of Tilt

Lateral Lift Point Load (F_{dl}): For lifting lug design, an out-of-plane lateral load, acting simultaneously with the in-plane load, of 5.0% of the maximum sling force (FSD). The lateral load shall be applied at center of the pinhole simultaneously with the sling force, and its associated moment shall also be considered as shown below.



$$F_{dl} = FSD \times \sin(\phi) \quad M_e = F_{dl} \times h$$

where " ϕ " is sling out of plane angle.



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Figure 14- Lateral Lift Forces

The overall impact of these factors may go up to 3.0 or more depending on environmental conditions, type of lifting / rigging, location of lifting, installation method and member or attachment. ASD (Allowable Stress Design) is typically used for Lifting analysis. The following is an example for calculating total lifting impact factor. (For reference only).

Total Impact Factor for strength Load combination, if D is the total Dead Load of module. The following impact factor will be applied on "D", while doing the lifting analysis.

$$\text{Impact Factor} = f_1 \times f_2 \times f_3 \times f_4 \times f_5 \times f_6$$

Weight contingency factor (f_1) 1.1 shall be used if the module weight is from calculation or 3D model.

It is recommended to use CoG envelope in lifting analysis, otherwise, center of gravity factor (f_2) 1.1 shall be used for not considering COG envelope. MWS agreement is required if CoG envelop factor is used.

Dynamic Amplification factor (f_3): The factor by which the weight is multiplied, to account for accelerations and impacts during lifting. A factor of 1.1 can be used for module weighing <100 tonnes, where land-crane is used and 1.12 where vessel's crane is used.

Skew factor (f_4): factor of 1.1 can be used for basic lift design, and with lift system incorporating spreader bars using matched pair of slings.

To allow for rotation of load about its vertical axis, a yaw factor (f_5) of 1.05 shall be adopted.

The Member Consequence Factor (MCF), f_6 , is as described above in lift factors.

As per above consideration,

$$\begin{aligned} \text{Impact Factor} &= 1.1 \times 1.1 \times 1.1 \times 1.1 \times 1.05 \times \text{MCF} \text{ where land-crane is used} \\ &= \text{MCF} \times 1.5 \end{aligned}$$

Lifting analysis for "Lift-On, Lift-Off" would be typically ASD (Allowable Stress Design) with an allowable stress of $0.6F_y$. Therefore, using the impact factors calculated above with consideration of CoG envelope and enforcement displacement for tilt in analysis model for a tandem lift shall be compared against $0.6F_y$

For lift lug or pad eyes, MCF = 1.3 (Highlighted with Red in below snap)

$$\text{Impact Factor} = 1.3 \times 1.45 = 1.89$$

For members attached to lift lug or pad eyes, MCF = 1.15 (Highlighted with Blue in below snap)

$$\text{Impact Factor} = 1.15 \times 1.45 = 1.67$$

For other members, MCF = 1.00 (Highlighted with Yellow in below snap)

$$\text{Impact Factor} = 1.00 \times 1.45 = 1.45$$



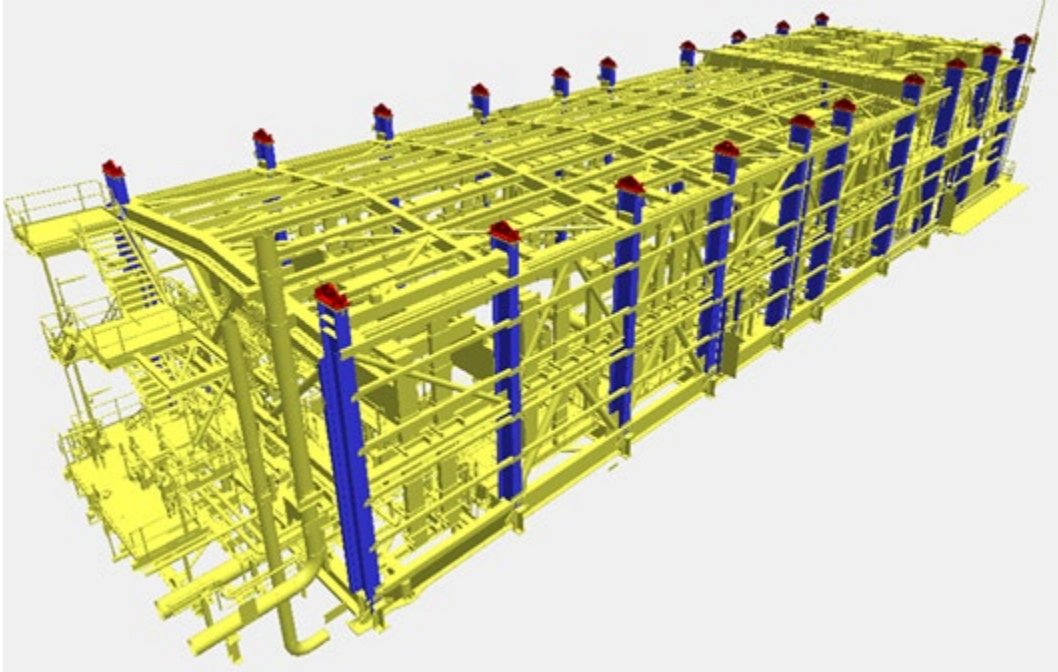
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Figure 15- Determining Member Consequence Factors

The following are few points which need to be taken care while producing engineering documents / drawings / reports related to lifting operations.

- Each module is structurally independent for lifting (either with a jacking system, or with a crane) and transportation, and is designed specifically for the lifting and transportation approaches used from the module fabrication yard to its final location at the jobsite.
- Module accessories for lifting, bracing, and transportation, and any additional structural steel required in a module due to shipping loads will be designed and supplied as a part of the engineering of the module. Structural drawings indicate all transport beams, lifting, and jacking locations, as required.
- The lifting configuration coming from the lifting contractor including information on sling and spreader beam dimensions, cross section and materials should be acquired and used in creating the mathematical model for analysis. Hence, the contractor engineering works should be consistent with module structural engineering schedule.
- Modules identified to be lifted with a crane, will have a minimum of four (4) pick points located on the drawings, and designed such that deflections and stresses during module handling are within allowable limits.
- Module lift analysis should consider rigging with the pin boundary condition at the crane hook.
- If the piperack modules are designed to have air coolers pre-installed, consideration should be given to bent spacing and lifting point locations.



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- Make every effort to keep all lifting points at the same elevation with consistent spacing (rigging gear is purchased in standard lengths and it is easier to make field adjustments with consistent lug spacing assuring the rigging at the lifting points are taking their anticipated loads).
- The rigging crews are more productive if they do not need to change the rigging gear between module lifts, especially for piperacks. Try to standardize the number of lift points and the distance between lift points.
- Designing the top transverse beam on a piperack module to take the compressive force from lifting will reduce the need for transverse lifting beams. Elimination of lifting beams increases craft productivity, since they have fewer components in the rigging hook up and handling of the gear.
- Permanent structural lifting lugs and pad eyes are preferred over removable lifting lugs, and eliminate extra handling and costs associated with installation and removal of the lugs.
- Removable and / or reusable lifting lugs may be used where it's impractical to integrate permanent pad eyes or lifting lugs, or to facilitate the stacking of multiple modules. Whenever possible, standardize lug design, with design drawings indicating lifting lug position, type, and Safe Weight Limit.
- If lifting lugs are being reused, fatigue design needs to be considered, in addition to inspection of lugs prior to each use.

We understand that lifting operation is a very extensive process. The Structural Engineer plays a very vital role in this process, by checking the strength and stability of the module structure globally and by passing the forces / reactions to Logistics for further action. The Structural Engineer needs to be aware of all project & stakeholder requirements.

6.9.4 Grillage Design:

Refer to Attachment 10 for the recommended grillage design methodology. Grillage may be designed by others.

7.0 RESOURCES

Not Used

8.0 REFERENCES

Guidelines

Guideline 000.200.0880	Weight Control Management
Guideline 000.250.1220	Piping Stress Engineering – Sea Transportation Analysis for Modules
Guideline 000.215.0766	Large and Very Large Modules – CSA Efficiency and Consistency

Non-Fluor Documents

AISC 360-16	Specification for Structural Steel Buildings
AISC Steel Construction Manual 2019	



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DNVGL-ST-N001 (2016-06)	Marine Operations and Marine Warranty
ISO 19902 :2020	Petroleum and natural gas industries-Fixed steel offshore structures

9.0 ATTACHMENTS / ADDENDA
Attachments

Attachment 01	Marine Warranty Surveyor Activities During Project Execution
Attachment 02	WA-OCEAN Spreadsheet (MS Excel attachment)
Attachment 03	Procedure to Design for Ocean Transportation
Attachment 04	Definition of Axis Orientations and Positive Directions
Attachment 05	Derivation of Component Load Combinations
Attachment 06	Derivation of Code Load Combinations
Attachment 07	Derivation of Center of Gravity Shift Equations
Attachment 08	Definition of Wave Induced Loads
Attachment 09	Derivation of Hog and Sag Deformations
Attachment 10	Grillage and Sea Fastening Design

Addenda

Not Used